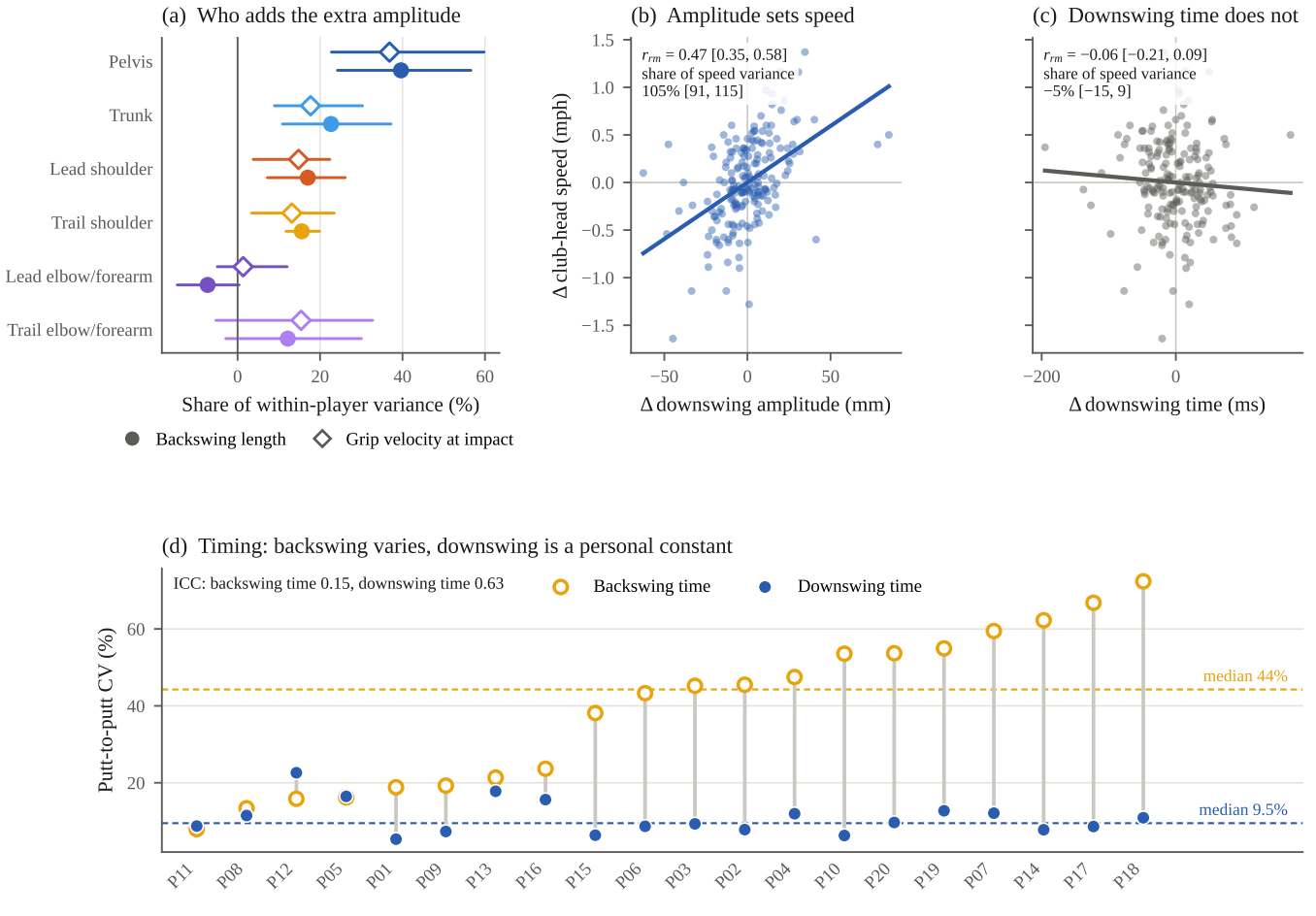
(a) Who adds the extra amplitude
Pelvis
Trunk
Lead shoulder
Trail shoulder
Lead elbow/forearm
Trail elbow/forearm
0
20
40
60
Share of within-player variance (%)
Backswing length
Grip velocity at impact
(b) Amplitude sets speed
r_{rm} = 0.47 [0.35, 0.58]
share of speed variance
105% [91, 115]
Δ club-head speed (mph)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−50
0
50
Δ downswing amplitude (mm)
(c) Downswing time does not
r_{rm} = −0.06 [−0.21, 0.09]
share of speed variance
−5% [−15, 9]
−200
0
Δ downswing time (ms)
(d) Timing: backswing varies, downswing is a personal constant
ICC: backswing time 0.15, downswing time 0.63
Backswing time
Downswing time
Putt-to-putt CV (%)
60
40
20
median 44%
median 9.5%
P11
P08
P12
P05
P01
P09
P13
P16
P15
P06
P03
P02
P04
P10
P20
P19
P07
P14
P17
P18